

MULTIPLICATION AND DIVISION — TIMES TABLES TO 12×12,

Name: _____

Date: _____

LO: I can identify factor pairs of a number and use the commutative law of multiplication.

Factors and commutativity

A **factor** is a whole number that divides exactly into another number. **Factor pairs** are two factors that multiply together to give a particular product.

Commutativity means the order of multiplication does not change the answer: $a \times b = b \times a$. This makes it easier to learn times tables because every fact gives you two results.

Quick examples

●	$1 \times 12 = 12$	factor pair of 12
●	$3 \times 4 = 4 \times 3 = 12$	commutativity shown
●	$2 \times 6 = 12$	another factor pair of 12
●	$5 \times 3 = 12$	$5 \times 3 = 15$, not 12
●	$3 + 4 = 3 \times 4$	addition and multiplication are different operations

Key idea

To find all factor pairs, start from 1 and work up; stop when the factors begin to repeat.

$$24 = 1 \times 24 = 2 \times 12 = 3 \times 8 = 4 \times 6$$

Four factor pairs of 24

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 – listing factor pairs

Find all the factor pairs of 18.

Start: 1×18

Check 2: 2×9

Check 3: 3×6 ; 4 does not divide 18 exactly, so stop

Factor pairs: (1, 18), (2, 9), (3, 6)

Work systematically from 1 upwards until factors repeat.

Example 2 – using commutativity

Use the commutative law to write two multiplication facts from $7 \times 8 = 56$.

First fact: $7 \times 8 = 56$

Swap: $8 \times 7 = 56$

Both give the same product

$7 \times 8 = 56$ and $8 \times 7 = 56$

The commutative law means $a \times b = b \times a$ always.

Example 3 – identifying missing factors

Find the missing number: $__ \times 9 = 36$.

We need a number that multiplies by 9 to give 36

$$36 \div 9 = 4$$

$$\text{Check: } 4 \times 9 = 36$$

The missing number is 4

Division is the inverse of multiplication; use it to find a missing factor.

Example 4 – factor pairs of a larger number

List all factor pairs of 36.

$1 \times 36, 2 \times 18, 3 \times 12, 4 \times 9, 6 \times 6$

Stop at 6×6 because 7 does not divide 36 exactly

There are 5 factor pairs (including the square root pair)

Factor pairs: (1, 36), (2, 18), (3, 12), (4, 9), (6, 6)

36 is a square number so one pair has the same factor repeated.

YOUR TURN – PRACTICE (deliberate practice)

1. Finding factor pairs

List all the factor pairs of each number.

- a) 12
- b) 20
- c) 30
- d) 16

2. Using commutativity

Write two multiplication facts for each product.

- a) Product = 42 (use 6 and 7)
- b) Product = 54 (use 6 and 9)
- c) Product = 72 (use 8 and 9)
- d) Product = 48 (use 6 and 8)

3. Missing factors

Find the missing number in each multiplication.

- a) $__ \times 7 = 63$
- b) $8 \times __ = 96$
- c) $__ \times 11 = 132$
- d) $4 \times __ = 48$

4. Factors of larger numbers

List all factor pairs. State how many factors each number has.

- a) 48
- b) 60
- c) 25
- d) 100

5. True or false

State whether each statement is true or false. Explain your answer.

- a) 7 is a factor of 49
- b) $5 \times 8 = 8 \times 5$ because multiplication is commutative
- c) The factor pairs of 9 are (1,9) and (3,3)
- d) 15 has exactly 4 factors
- e) Division is commutative

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) $3 \times 5 = 5 \times 3$ because multiplication is commutative

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Correct answer: _____

Reason: _____

b) $12 \div 4 = 4 \div 12$ because division is also commutative

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Correct answer: _____

Reason: _____

c) Factor pairs of 24 include (3, 8) and (4, 6)

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Correct answer: _____

Reason: _____

d) 7 is a factor of 50

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Correct answer: _____

Reason: _____

e) Every number has at least two factors: 1 and itself

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Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) A teacher has **36 exercise books** to arrange into equal piles. List all the ways she can arrange them into equal piles (each pile having the same number of books). Which arrangement uses the most piles?

Expression: _____

Simplified: _____

b) Two numbers have a product of **144**. One number is **twice** the other. Use factor pairs to find the two numbers. Explain how commutativity helps you check your answer.

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

